DHAKA UNIVERSITY OF ENGINEERING & TECHNOLOGY

DUET ADMISSION PREPARATION

C & C++ Programs

মৌলিক প্রোগ্রামসমূহ — পাশাপাশি তুলনা

C LANGUAGE | C++ LANGUAGE

109 Programs · Side by Side · Single Explanation

STUDENT INFORMATION

STUDENT NAME	STUDENT ID	BATCH

DEPARTMENT	POLYTECHNIC INSTITUTE	ACADEMIC YEAR

১. ক্যালকুলেটর - যোগ বিয়োগ গুণ ভাগ 1. CALCULATOR - SUM SUB MUL DIV

নমুনা ইনপুট/আউটপুট:

```
Enter two numbers: 10 3
Sum = 13.00
Subtraction = 7.00
Multiplication = 30.00
Division = 3.33
```

C

```c
#include <stdio.h>
int main() {
    float a, b;
    printf("Enter two numbers: ");
    scanf("%f %f", &a, &b);
    printf("Sum = %.2f\n", a + b);
    printf("Subtraction = %.2f\n", a - b);
    printf("Multiplication = %.2f\n", a * b);
    if (b != 0)
        printf("Division = %.2f\n", a / b);
    else
        printf("Division = Infinity\n");
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    float a, b;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    cout << "Sum = " << a + b << endl;
    cout << "Subtraction = " << a - b << endl;
    cout << "Multiplication = " << a * b << endl;
    if (b != 0)
        cout << "Division = " << a / b << endl;
    else
        cout << "Division = Infinity" << endl;
    return 0;
}
```

২ক. তিনটি সংখ্যার মধ্যে সবচেয়ে বড় 2a. LARGEST AMONG 3 NUMBERS

C

```c
#include <stdio.h>
int main() {
    int a, b, c;
    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);
    if (a > b && a > c) {
        printf("%d is the largest.\n", a);
    }
    else if (b > c) {
        printf("%d is the largest.\n", b);
    }
    else {
        printf("%d is the largest.\n", c);
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int a, b, c;
    cout << "Enter three numbers: ";
    cin >> a >> b >> c;
    if (a > b && a > c) {
        cout << a << " is the largest." << endl;
    }
    else if (b > c) {
        cout << b << " is the largest." << endl;
    }
    else {
        cout << c << " is the largest." << endl;
    }
    return 0;
}
```

২খ. তিনটি সংখ্যার মধ্যে সবচেয়ে ছোট 2b. SMALLEST AMONG 3 NUMBERS

C

```c
#include <stdio.h>
int main() {
    int a, b, c;
    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);
    if (a < b && a < c) {
        printf("%d is the smallest.\n", a);
    }
    else if (b < c) {
        printf("%d is the smallest.\n", b);
    }
    else {
        printf("%d is the smallest.\n", c);
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int a, b, c;
    cout << "Enter three numbers: ";
    cin >> a >> b >> c;
    if (a < b && a < c) {
        cout << a << " is the smallest." << endl;
    }
    else if (b < c) {
        cout << b << " is the smallest." << endl;
    }
    else {
        cout << c << " is the smallest." << endl;
    }
    return 0;
}
```

২গ. বড় ও ছোট উভয় (একসাথে) 2c. LARGEST & SMALLEST (BOTH)

C

```c
#include <stdio.h>
int main() {
    int a, b, c;
    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);
    if (a > b && a > c) {
        printf("%d is the largest.\n", a);
    }
    else if (b > c) {
        printf("%d is the largest.\n", b);
    }
    else {
        printf("%d is the largest.\n", c);
    }

    if (a < b && a < c) {
        printf("%d is the smallest.\n", a);
    }
    else if (b < c) {
        printf("%d is the smallest.\n", b);
    }
    else {
        printf("%d is the smallest.\n", c);
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int a, b, c;
    cout << "Enter three numbers: ";
    cin >> a >> b >> c;
    if (a > b && a > c) {
        cout << a << " is the largest." << endl;
    }
    else if (b > c) {
        cout << b << " is the largest." << endl;
    }
    else {
        cout << c << " is the largest." << endl;
    }
    if (a < b && a < c) {
        cout << a << " is the smallest." << endl;
    }
    else if (b < c) {
        cout << b << " is the smallest." << endl;
    }
    else {
        cout << c << " is the smallest." << endl;
    }
    return 0;
}
```

৩. তিনটি সংখ্যার গড় 3. AVERAGE OF 3 NUMBERS

গড় (Average) সূত্র:
গড় = (সংখ্যাগুলোর যোগফল) ÷ (সংখ্যার পরিমাণ)
উদাহরণ: (10 + 20 + 30) ÷ 3 = 60 ÷ 3 = 20.00

C

```c
#include <stdio.h>
int main() {
    float a, b, c, avg;
    printf("Enter three numbers: ");
    scanf("%f %f %f", &a, &b, &c);
    avg = (a + b + c) / 3;
    printf("Average = %.2f\n", avg);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    float a, b, c, avg;
    cout << "Enter three numbers: ";
    cin >> a >> b >> c;
    avg = (a + b + c) / 3;
    cout << "Average = " << avg << endl;
    return 0;
}
```

৪ক. সমকোণী ত্রিভুজের ক্ষেত্রফল 4a. RIGHT ANGLE TRIANGLE AREA

সমকোণী ত্রিভুজের ক্ষেত্রফল সূত্র:
ক্ষেত্রফল = (ভূমি × উচ্চতা) ÷ 2
উদাহরণ: ভূমি = 6, উচ্চতা = 4 → ক্ষেত্রফল = (6 × 4) ÷ 2 = 12.00

C

```c
#include <stdio.h>
int main() {
    float base, height, area;
    printf("Enter base and height: ");
    scanf("%f %f", &base, &height);
    area = (base * height) / 2;
    printf("Right Angle Triangle Area = %.2f\n", area);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    float base, height, area;
    cout << "Enter base and height: ";
    cin >> base >> height;
    area = (base * height) / 2;
    cout << "Right Angle Triangle Area = " << area << endl;
    return 0;
}
```

৪খ. সমবাহু ত্রিভুজের ক্ষেত্রফল 4b. EQUILATERAL TRIANGLE AREA

```
সমবাহু ত্রিভুজের ক্ষেত্রফলের সূত্র:
  ক্ষেত্রফল = (√3 / 4) × a²

  যেখানে a = তিনটি সমান বাহুর দৈর্ঘ্য
  √3 ≈ 1.7320508 (কোডে এই মান সরাসরি ব্যবহার করা হয়েছে,
  কারণ math.h ছাড়া sqrt() ফাংশন নেই)
```

C

```c
#include <stdio.h>
int main() {
    float a, area;
    printf("Enter side length: ");
    scanf("%f", &a);
    area = (1.7320508 / 4) * a * a;
    printf("Equilateral Triangle Area = %.2f\n", area);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    float a, area;
    cout << "Enter side length: ";
    cin >> a;
    area = (1.7320508 / 4) * a * a;
    cout << "Equilateral Triangle Area = " << area << endl;
    return 0;
}
```

৪গ. বিষমবাহু ত্রিভুজের ক্ষেত্রফল 4c. SCALENE TRIANGLE AREA

```
বিষমবাহু ত্রিভুজের ক্ষেত্রফল — হেরনের সূত্র:
  s = (a + b + c) / 2      ← অর্ধ-পরিসীমা (semi-perimeter)
  ক্ষেত্রফল = √(s × (s-a) × (s-b) × (s-c))

  math.h ছাড়া sqrt() নেই, তাই Newton's method দিয়ে হাতে বর্গমূল বের করা হয়েছে:
  guess = (guess + x / guess) / 2   ← ২০ বার চালালে নির্ভুল উত্তর আসে।
```

C

```c
#include <stdio.h>
int main() {
    float a, b, c, s, x, area;
    printf("Enter sides a, b, c: ");
    scanf("%f %f %f", &a, &b, &c);
    if (a <= 0 || b <= 0 || c <= 0 || a + b <= c || a + c <= b |
| b + c <= a) {
        printf("Error: invalid triangle sides.\n");
        return 1;
    }
    s = (a + b + c) / 2;
    x = s * (s - a) * (s - b) * (s - c);
    float guess = x;
    for (int i = 0; i < 20; i++) {
    guess = (guess + x / guess) / 2;
    }
    area = guess;
    printf("Scalene Triangle Area = %.2f\n", area);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    float a, b, c, s, x, area;
    cout << "Enter sides a, b, c: ";
    cin >> a >> b >> c;
    if (a <= 0 || b <= 0 || c <= 0 || a + b <= c || a + c <= b |
| b + c <= a) {
        cout << "Error: invalid triangle sides." << endl;
        return 1;
    }
    s = (a + b + c) / 2;
    x = s * (s - a) * (s - b) * (s - c);
    float guess = x;
    for (int i = 0; i < 20; i++) {
    guess = (guess + x / guess) / 2;
    }
    area = guess;
    cout << "Scalene Triangle Area = " << area << endl;
    return 0;
}
```

৫. বৃত্তের ক্ষেত্রফল ও পরিধি 5. CIRCLE AREA & CIRCUMFERENCE

```
বৃত্তের সূত্র:
  ক্ষেত্রফল     = π × r²        (π ≈ 3.1416)
  পরিধি (circumference) = 2 × π × r

  π (পাই) কী?  যেকোনো বৃত্তের পরিধি ÷ ব্যাস = সবসময় ≈ 3.14159...
  এটি একটি অমূলদ সংখ্যা — শেষ হয় না, পুনরাবৃত্তিও হয় না।
  কোডে 3.1416 ব্যবহার করা হয় — এটি যথেষ্ট নির্ভুল।
```

C

```c
#include <stdio.h>
int main() {
    float r, pi = 3.1416, area, circ;
    printf("Enter radius: ");
    scanf("%f", &r);
    area = pi * r * r;
    circ = 2 * pi * r;
    printf("Area = %.4f Sq M\n", area);
    printf("Circumference = %.4f M\n", circ);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    float r, pi = 3.1416, area, circ;
    cout << "Enter radius: ";
    cin >> r;
    area = pi * r * r;
    circ = 2 * pi * r;
    cout << "Area = " << area << " Square Meter" << endl;
    cout << "Circumference = " << circ << " Meter" << endl;
    return 0;
}
```

৬. ধনাত্মক না ঋণাত্মক সংখ্যা 6. POSITIVE OR NEGATIVE NUMBER

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);
    if (n > 0) {
        printf("%d is Positive.\n", n);
    }
    else if (n < 0) {
        printf("%d is Negative.\n", n);
    }
    else {
        printf("The number is Zero.\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter a number: ";
    cin >> n;
    if (n > 0) {
        cout << n << " is Positive." << endl;
    }
    else if (n < 0) {
        cout << n << " is Negative." << endl;
    }
    else {
        cout << "The number is Zero." << endl;
    }
    return 0;
}
```

৭. বিজোড় না জোড় 7. ODD OR EVEN

```
বিজোড়/জোড় কীভাবে বোঝা যায়?
  % চিহ্ন মানে ভাগশেষ (remainder)।
  n % 2 == 0  → ভাগশেষ শূন্য → জোড় (Even)
  n % 2 == 1  → ভাগশেষ এক  → বিজোড় (Odd)
  উদাহরণ: 7 % 2 = 1 → Odd  |  8 % 2 = 0 → Even
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);
    if (n % 2 == 0) {
        printf("%d is Even.\n", n);
    }
    else {
        printf("%d is Odd.\n", n);
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter a number: ";
    cin >> n;
    if (n % 2 == 0) {
        cout << n << " is Even." << endl;
    }
    else {
        cout << n << " is Odd." << endl;
    }
    return 0;
}
```

৮. অধিবর্ষ (লিপ ইয়ার) 8. LEAP YEAR

```
অধিবর্ষ (Leap Year) কেন?
পৃথিবী সূর্যকে একবার প্রদক্ষিণ করতে ঠিক ৩৬৫ দিন নয়, বরং ৩৬৫ দিন ৫ ঘণ্টা ৪৮ মিনিট ৪৫ সেকেন্ড লাগে।
প্রতি বছর প্রায় ৬ ঘণ্টা বাকি থাকে → ৪ বছরে ≈ ২৪ ঘণ্টা = ১ দিন।
তাই প্রতি ৪ বছরে ফেব্রুয়ারিতে ১ দিন যোগ করা হয় → ফেব্রুয়ারি ২৯ দিনের হয়।

কিন্তু ৬ ঘণ্টা নয়, আসলে ৫ ঘণ্টা ৪৮ মিনিট — তাই প্রতি ৪ বছরে একটু বেশি যোগ হয়।
১০০ বছরে মোট ৪৪ মিনিট বেশি হয় → তাই ১০০-এ বিভাজ্য বছরগুলো leap year নয়।
তবু ৪০০ বছরে ৩ দিন ঘাটতি পড়ে → তাই ৪০০-এ বিভাজ্য হলে আবার leap year।

নিয়ম:
  ১. year % 4 == 0        → সম্ভব leap year
  ২. year % 100 == 0      → leap year নয় (বাদ)
  ৩. year % 400 == 0      → আবার leap year (ব্যতিক্রম)
```

C

```c
#include <stdio.h>
int main() {
    int year;
    printf("Enter a year: ");
    scanf("%d", &year);
    if ((year % 4 == 0 && year % 100 != 0) || year % 400 == 0)
        printf("%d is a Leap Year.\n", year);
    else
        printf("%d is not a Leap Year.\n", year);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int year;
    cout << "Enter a year: ";
    cin >> year;
    if ((year % 4 == 0 && year % 100 != 0) || year % 400 == 0)
        cout << year << " is a Leap Year." << endl;
    else
        cout << year << " is not a Leap Year." << endl;
    return 0;
}
```

৯ক. সেলসিয়াস থেকে ফারেনহাইট 9a. CELSIUS TO FAHRENHEIT

```
সেলসিয়াস → ফারেনহাইট রূপান্তর সূত্র:
  F = (C × 9/5) + 32
  উদাহরণ: 0°C → (0 × 9/5) + 32 = 32°F   (পানি জমার বিন্দু)
          100°C → (100 × 9/5) + 32 = 212°F  (পানি ফুটার বিন্দু)
```

C

```c
#include <stdio.h>
int main() {
    float c, f;
    printf("Enter temperature in Celsius: ");
    scanf("%f", &c);
    f = (c * 9 / 5) + 32;
    printf("%.2f C = %.2f F\n", c, f);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    float c, f;
    cout << "Enter temperature in Celsius: ";
    cin >> c;
    f = (c * 9 / 5) + 32;
    cout << c << " C = " << f << " F" << endl;
    return 0;
}
```

৯খ. ফারেনহাইট থেকে সেলসিয়াস 9b. FAHRENHEIT TO CELSIUS

```
ফারেনহাইট → সেলসিয়াস রূপান্তর সূত্র:
  C = (F - 32) × 5/9
  উদাহরণ: 32°F → (32 - 32) × 5/9 = 0°C
          98.6°F → (98.6 - 32) × 5/9 = 37°C  (মানব শরীরের তাপমাত্রা)
```

C

```c
#include <stdio.h>
int main() {
    float f, c;
    printf("Enter temperature in Fahrenheit: ");
    scanf("%f", &f);
    c = (f - 32) * 5 / 9;
    printf("%.2f F = %.2f C\n", f, c);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    float f, c;
    cout << "Enter temperature in Fahrenheit: ";
    cin >> f;
    c = (f - 32) * 5 / 9;
    cout << f << " F = " << c << " C" << endl;
    return 0;
}
```

১০ক. বড় হাতে না ছোট হাতের অক্ষর (রেঞ্জ দিয়ে) 10a. UPPERCASE OR LOWERCASE (RANGE)

```
ASCII রেঞ্জ দিয়ে বড়/ছোট হাত যাচাই:
  বড় হাত (A-Z): 'A' = 65, 'Z' = 90
  ছোট হাত (a-z): 'a' = 97, 'z' = 122
  শর্ত: ch ≥ 'A' && ch ≤ 'Z'  →  বড় হাত
        ch ≥ 'a' && ch ≤ 'z'  →  ছোট হাত
  (অক্ষর সরাসরি তুলনা করা যায় কারণ অক্ষর = সংখ্যা)
```

C

```c
#include <stdio.h>
int main() {
    char ch;
    printf("Enter a letter: ");
    scanf(" %c", &ch);
    if (ch >= 97 && ch <= 122) {
        printf("%c is Lowercase.\n", ch);
    }
    else if (ch >= 65 && ch <= 90) {
        printf("%c is Uppercase.\n", ch);
    }
    else {
        printf("%c is not a letter.\n", ch);
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    char ch;
    cout << "Enter a letter: ";
    cin >> ch;
    if (ch >= 97 && ch <= 122) {
        cout << ch << " is Lowercase." << endl;
    }
    else if (ch >= 65 && ch <= 90) {
        cout << ch << " is Uppercase." << endl;
    }
    else {
        cout << ch << " is not a letter." << endl;
    }
    return 0;
}
```

১০খ. বড় হাতে না ছোট হাতের অক্ষর (ASCII দিয়ে) 10b. UPPERCASE OR LOWERCASE (ASCII)

```
ASCII কী?
কম্পিউটার শুধু সংখ্যা বোঝে। তাই প্রতিটি অক্ষরকে একটি সংখ্যায় রূপান্তর করা হয়।
এই সংখ্যা-তালিকার নাম ASCII (American Standard Code for Information Interchange)।

  'A' = 65      'a' = 97
  'B' = 66      'b' = 98
  'C' = 67      'c' = 99
  ...           ...
  'Z' = 90      'z' = 122

বড় হাত (Uppercase): 65 থেকে 90  → if (ch ≥ 65 && ch ≤ 90)
ছোট হাত (Lowercase): 97 থেকে 122 → if (ch ≥ 97 && ch ≤ 122)

কেন 65 থেকে শুরু? এটা ঐতিহাসিক — 1963 সালে ASCII ডিজাইন করার সময়
0-64 রাখা হয়েছিল বিশেষ চিহ্ন ও নিয়ন্ত্রণ কোডের জন্য।
```

C

```c
#include <stdio.h>
int main() {
    char ch;
    printf("Enter a letter: ");
    scanf(" %c", &ch);
    if (ch >= 65 && ch <= 90) {
        printf("%c is Uppercase. ASCII = %d\n", ch, ch);
    }
    else if (ch >= 97 && ch <= 122) {
        printf("%c is Lowercase. ASCII = %d\n", ch, ch);
    }
    else {
        printf("%c is not a letter.\n", ch);
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    char ch;
    cout << "Enter a letter: ";
    cin >> ch;
    if (ch >= 65 && ch <= 90) {
        cout << ch << " is Uppercase. ASCII = " << (int)ch << en
dl;
    }
    else if (ch >= 97 && ch <= 122) {
        cout << ch << " is Lowercase. ASCII = " << (int)ch << en
dl;
    }
    else {
        cout << ch << " is not a letter." << endl;
    }
    return 0;
}
```

১১ক. স্বরবর্ণ না ব্যঞ্জনবর্ণ - if দিয়ে 11a. VOWEL OR CONSONANT - IF

```
স্বরবর্ণ (Vowel): a, e, i, o, u  (বড় ও ছোট উভয়)
ব্যঞ্জনবর্ণ (Consonant): বাকি সব অক্ষর।
if দিয়ে প্রতিটি সম্ভাব্য মান লিখতে হয়:
  if (ch='a' || ch='e' || ch='i' || ch='o' || ch='u' || ch='A' ...)
```

C

```c
#include <stdio.h>
int main() {
    char ch;
    printf("Enter a character: ");
    scanf(" %c", &ch);
    if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch =
= 'u' ||
        ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch =
= 'U')
        printf("%c is a Vowel.\n", ch);
    else
        printf("%c is a Consonant.\n", ch);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    char ch;
    cout << "Enter a character: ";
    cin >> ch;
    if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch =
= 'u' ||
        ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch =
= 'U')
        cout << ch << " is a Vowel." << endl;
    else
        cout << ch << " is a Consonant." << endl;
    return 0;
}
```

১১খ. স্বরবর্ণ না ব্যঞ্জনবর্ণ - switch দিয়ে 11b. VOWEL OR CONSONANT - SWITCH

switch দিয়ে একই কাজ আরও পরিষ্কারভাবে লেখা যায়।
case 'a': case 'e': case 'i': ... — একাধিক case একসাথে লেখা যায়,
break না থাকলে পরবর্তী case-এও চলে যায়। এখানে সেটাই কাজে লাগানো হয়েছে।

C

```c
#include <stdio.h>
int main() {
    char ch;
    printf("Enter a character: ");
    scanf(" %c", &ch);
    switch (ch) {
        case 'a': case 'e': case 'i': case 'o': case 'u':
        case 'A': case 'E': case 'I': case 'O': case 'U':
            printf("%c is a Vowel.\n", ch);
            break;
        default:
            printf("%c is a Consonant.\n", ch);
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    char ch;
    cout << "Enter a character: ";
    cin >> ch;
    switch (ch) {
        case 'a': case 'e': case 'i': case 'o': case 'u':
        case 'A': case 'E': case 'I': case 'O': case 'U':
            cout << ch << " is a Vowel." << endl;
            break;
        default:
            cout << ch << " is a Consonant." << endl;
    }
    return 0;
}
```

§ LOOPS — REPETITION

১২ক. ১০ বার প্রিন্ট - for লুপ দিয়ে 12a. PRINT 10 TIMES - FOR LOOP

তিন ধরনের লুপের পার্থক্য:
for → শুরু, শর্ত, পরিবর্তন — তিনটি একলাইনে। গণনার জন্য সবচেয়ে উপযুক্ত।
while → আগে শর্ত দেখে, তারপর চালায়। শর্ত মিথ্যা হলে একবারও চলে না।
do-while → আগে একবার চালায়, তারপর শর্ত দেখে। অন্তত একবার চলবেই।

C

```c
#include <stdio.h>
int main() {
    for (int i = 1; i <= 10; i++) {
    printf("Bangladesh\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 10; i++) {
    cout << "Bangladesh" << endl;
    }
    return 0;
}
```

১২খ. ১০ বার প্রিন্ট - while লুপ দিয়ে 12b. PRINT 10 TIMES - WHILE LOOP

C

```c
#include <stdio.h>
int main() {
    int i = 1;
    while (i <= 10) {
    printf("Bangladesh\n");
    i++;
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int i = 1;
    while (i <= 10) {
    cout << "Bangladesh" << endl;
    i++;
    }
    return 0;
}
```

১২গ. ১০ বার প্রিন্ট - do-while দিয়ে 12c. PRINT 10 TIMES - DO-WHILE

C

```c
#include <stdio.h>
int main() {
    int i = 1;
    do {
    printf("Bangladesh\n");
    i++;
    } while (i <= 10);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int i = 1;
    do {
    cout << "Bangladesh" << endl;
    i++;
    } while (i <= 10);
    return 0;
}
```

১২ঘ. N বার টেক্সট প্রিন্ট - for দিয়ে 12d. PRINT TEXT N TIMES - FOR

C

```c
#include <stdio.h>
int main() {
    char text[100];
    int n;
    printf("Enter text: ");
    scanf(" %[^\n]", text);
    printf("How many times? ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
    printf("%s\n", text);
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
#include<string>
using namespace std;
int main() {
    string text;
    int n;
    cout << "Enter text: ";
    getline(cin >> ws, text);
    cout << "How many times? ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
    cout << text << endl;
    }
    return 0;
}
```

১২ঙ. N বার টেক্সট প্রিন্ট - while দিয়ে 12e. PRINT TEXT N TIMES - WHILE

C

```c
#include <stdio.h>
int main() {
    char text[100];
    int n, i = 1;
    printf("Enter text: ");
    scanf(" %[^\n]", text);
    printf("How many times? ");
    scanf("%d", &n);
    while (i <= n) {
    printf("%s\n", text);
    i++;
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
#include<string>
using namespace std;
int main() {
    string text;
    int n, i = 1;
    cout << "Enter text: ";
    getline(cin >> ws, text);
    cout << "How many times? ";
    cin >> n;
    while (i <= n) {
    cout << text << endl;
    i++;
    }
    return 0;
}
```

১২চ. N বার টেক্সট প্রিন্ট - do-while দিয়ে 12f. PRINT TEXT N TIMES - DO-WHILE

C

```c
#include <stdio.h>
int main() {
    char text[100];
    int n, i = 1;
    printf("Enter text: ");
    scanf(" %[^\n]", text);
    printf("How many times? ");
    scanf("%d", &n);
    do {
    printf("%s\n", text);
    i++;
    } while (i <= n);
    return 0;
}
```

C++

```cpp
#include<iostream>
#include<string>
using namespace std;
int main() {
    string text;
    int n, i = 1;
    cout << "Enter text: ";
    getline(cin >> ws, text);
    cout << "How many times? ";
    cin >> n;
    do {
    cout << text << endl;
    i++;
    } while (i <= n);
    return 0;
}
```

১৩ক. ১ থেকে ১০ পর্যন্ত প্রিন্ট 13a. PRINT 1 TO 10

C

```c
#include <stdio.h>
int main() {
    for (int i = 1; i <= 10; i++) {
    printf("%d ", i);
    }
    printf("\n");
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 10; i++) {
    cout << i << " ";
    }
    cout << endl;
    return 0;
}
```

১৩খ. ১ থেকে N পর্যন্ত প্রিন্ট 13b. PRINT 1 TO N

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
    printf("%d ", i);
    }
    printf("\n");
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
    cout << i << " ";
    }
    cout << endl;
    return 0;
}
```

১৪ক. ১ থেকে ১০ পর্যন্ত যোগফল 14a. SUM OF 1 TO 10

```
১ থেকে ১০ পর্যন্ত যোগফল:
  1 + 2 + 3 + ... + 10 = 55
  সূত্র: n(n+1)/2 = 10×11/2 = 55  (লুপ না চালিয়েও পাওয়া যায়)
```

C

```c
#include <stdio.h>
int main() {
    int sum = 0;
    for (int i = 1; i <= 10; i++) {
    sum = sum + i;
    }
    printf("Sum = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int sum = 0;
    for (int i = 1; i <= 10; i++) {
    sum = sum + i;
    }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

§ NUMBER THEORY

১৪খ. ১ থেকে N পর্যন্ত যোগফল 14b. SUM OF 1 TO N

C

```c
#include <stdio.h>
int main() {
    int n, sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
    sum = sum + i;
    }
    printf("Sum = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n, sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
    sum = sum + i;
    }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

১৫. মৌলিক সংখ্যা যাচাই 15. PRIME NUMBER CHECK

```
মৌলিক সংখ্যা (Prime) কী?
যে সংখ্যাকে শুধু ১ এবং নিজেকে দিয়ে ভাগ যায়: 2, 3, 5, 7, 11, 13...

কেন i ≤ n/2 পর্যন্ত দেখা যথেষ্ট?
  n=36 এর গুণনীয়ক: 2×18, 3×12, 4×9, 6×6 — বড় অংশটি n/2 এর বাইরে।
  তাই n/2 পর্যন্ত খুঁজলেই সব factor পাওয়া যায়।

বিশেষ নিয়ম: ১ মৌলিক নয়, ২ একমাত্র জোড় মৌলিক সংখ্যা।
```

C

```c
#include <stdio.h>
int main() {
    int n, isPrime = 1;
    printf("Enter a number: ");
    scanf("%d", &n);
    if (n <= 1) {
        isPrime = 0;
    }
    for (int i = 2; i * i <= n; i++) {
    if (n % i == 0) {
    isPrime = 0;
    break;
    }
    }
    if (isPrime) {
        printf("%d is a Prime number.\n", n);
    }
    else {
        printf("%d is not a Prime number.\n", n);
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n, isPrime = 1;
    cout << "Enter a number: ";
    cin >> n;
    if (n <= 1) {
        isPrime = 0;
    }
    for (int i = 2; i * i <= n; i++) {
    if (n % i == 0) {
    isPrime = 0;
    break;
    }
    }
    if (isPrime) {
        cout << n << " is a Prime number." << endl;
    }
    else {
        cout << n << " is not a Prime number." << endl;
    }
    return 0;
}
```

১৬. পরিপূর্ণ সংখ্যা যাচাই 16. PERFECT NUMBER CHECK

```
পরিপূর্ণ সংখ্যা কী?

একটি সংখ্যাকে পরিপূর্ণ (Perfect) বলা হয়
যদি তার নিজেকে বাদ দিয়ে সব গুণনীয়কের
যোগফল সংখ্যাটির সমান হয়।

উদাহরণ: 6
গুণনীয়ক: 1, 2, 3
1 + 2 + 3 = 6 ✓

উদাহরণ: 28
গুণনীয়ক: 1, 2, 4, 7, 14
1+2+4+7+14 = 28 ✓

পরিচিত পরিপূর্ণ সংখ্যা: 6, 28, 496, 8128
```

C

```c
#include <stdio.h>
int main() {
    int n, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &n);
    if (n <= 1) {
        printf("%d is not a Perfect number.\n", n);
        return 0;
    }
    for (int i = 1; i < n; i++)
    if (n % i == 0) {
    sum = sum + i;
    }
    if (sum == n) {
        printf("%d is a Perfect number.\n", n);
    }
    else {
        printf("%d is not a Perfect number.\n", n);
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n, sum = 0;
    cout << "Enter a number: ";
    cin >> n;
    if (n <= 1) {
        cout << n << " is not a Perfect number." << endl;
        return 0;
    }
    for (int i = 1; i < n; i++)
    if (n % i == 0) {
    sum = sum + i;
    }
    if (sum == n) {
        cout << n << " is a Perfect number." << endl;
    }
    else {
        cout << n << " is not a Perfect number." << endl;
    }
    return 0;
}
```

১৭. সংখ্যার অঙ্কসমূহের যোগফল 17. SUM OF DIGITS

```
অঙ্কসমূহের যোগফল কীভাবে?
  সংখ্যা: 1234
  1234 % 10 = 4  → শেষ অঙ্ক
  1234 / 10 = 123 → বাকি সংখ্যা
  একইভাবে: 123 → 3, 12 → 2, 1 → 1
  যোগফল: 4 + 3 + 2 + 1 = 10
```

C

```c
#include <stdio.h>
int main() {
    int n, sum = 0, temp;
    printf("Enter a number: ");
    scanf("%d", &n);
    temp = (n < 0) ? -n : n;
    while (temp != 0) {
    sum = sum + temp % 10;
    temp = temp / 10;
    }
    printf("Sum of digits = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n, sum = 0, temp;
    cout << "Enter a number: ";
    cin >> n;
    temp = (n < 0) ? -n : n;
    while (temp != 0) {
    sum = sum + temp % 10;
    temp = temp / 10;
    }
    cout << "Sum of digits of " << n << " = " << sum << endl;
    return 0;
}
```

১৮. সংখ্যা উল্টানো 18. REVERSE A NUMBER

```
সংখ্যা উল্টানোর পদ্ধতি:
  n = 1234,  rev = 0
  ধাপ ১: rev = 0×10 + 4 = 4,   n = 123
  ধাপ ২: rev = 4×10 + 3 = 43,  n = 12
  ধাপ ৩: rev = 43×10 + 2 = 432, n = 1
  ধাপ ৪: rev = 432×10 + 1 = 4321, n = 0  → শেষ
  ফলাফল: 4321
```

C

```c
#include <stdio.h>
int main() {
    int n, rev = 0;
    printf("Enter a number: ");
    scanf("%d", &n);
    while (n != 0) {
    rev = rev * 10 + n % 10;
    n = n / 10;
    }
    printf("Reversed number = %d\n", rev);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n, rev = 0;
    cout << "Enter a number: ";
    cin >> n;
    while (n != 0) {
    rev = rev * 10 + n % 10;
    n = n / 10;
    }
    cout << "Reversed number = " << rev << endl;
    return 0;
}
```

১৯. প্যালিন্ড্রোম সংখ্যা 19. PALINDROME NUMBER

উল্টালেও একই থাকে
এমন সংখ্যা।

```
121   → উল্টো → 121  ✓
12321 → উল্টো → 12321 ✓
123   → উল্টো → 321  ✗
```

C

```c
#include <stdio.h>
int main() {
    int n, rev = 0, temp;
    printf("Enter a number: ");
    scanf("%d", &n);
    if (n < 0) {
        printf("%d is not a Palindrome.\n", n);
        return 0;
    }
    temp = n;
    while (temp != 0) {
    rev = rev * 10 + temp % 10;
    temp = temp / 10;
    }
    if (rev == n) {
        printf("%d is a Palindrome.\n", n);
    }
    else {
        printf("%d is not a Palindrome.\n", n);
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n, rev = 0, temp;
    cout << "Enter a number: ";
    cin >> n;
    if (n < 0) {
        cout << n << " not Palindrome." << endl;
        return 0;
    }
    temp = n;
    while (temp != 0) {
    rev = rev * 10 + temp % 10;
    temp = temp / 10;
    }
    if (rev == n) {
        cout << n << " is Palindrome." << endl;
    }
    else {
        cout << n << " not Palindrome." << endl;
    }
    return 0;
}
```

২০. আর্মস্ট্রং সংখ্যা 20. ARMSTRONG NUMBER

প্রতিটি অঙ্কের n-তম ঘাতের যোগফল সংখ্যার সমান হলে আর্মস্ট্রং।

153: $1^3+5^3+3^3=153$ ✓
1634: 1 +6 +3 +4
=1634 ✓

তালিকা: 1,2,3,153, 370,371,407,1634...

C

```c
#include <stdio.h>
int main() {
    int n, temp, digits = 0, sum = 0, t;
    printf("Enter a number: ");
    scanf("%d", &n);
    temp = n;
    while (temp != 0) {
    digits++;
    temp = temp / 10;
    }
    temp = n;
    while (temp != 0) {
    t = temp % 10;
    int p = 1;
    for (int i = 0; i < digits; i++) {
    p = p * t;
    }
    sum = sum + p;
    temp = temp / 10;
    }
    if (sum == n) {
        printf("%d is Armstrong.\n", n);
    }
    else {
        printf("%d is not Armstrong.\n", n);
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n, temp, digits = 0, sum = 0, t;
    cout << "Enter a number: ";
    cin >> n;
    temp = n;
    while (temp != 0) {
    digits++;
    temp = temp / 10;
    }
    temp = n;
    while (temp != 0) {
    t = temp % 10;
    int p = 1;
    for (int i = 0; i < digits; i++) {
    p = p * t;
    }
    sum = sum + p;
    temp = temp / 10;
    }
    if (sum == n) {
        cout << n << " is Armstrong." << endl;
    }
    else {
        cout << n << " not Armstrong." << endl;
    }
    return 0;
}
```

২১. স্ট্রং সংখ্যা 21. STRONG NUMBER

প্রতিটি অঙ্কের ফ্যাক্টরিয়ালের যোগফল সংখ্যার সমান হলে স্ট্রং।

145: 1!+4!+5!
= 1+24+120 = 145 ✓

40585: 4!+0!+5!+8!+5!
= 40585 ✓

তালিকা: 1,2,145,40585

C

```c
#include <stdio.h>
int main() {
    int n, temp, sum = 0, digit, fact;
    printf("Enter a number: ");
    scanf("%d", &n);
    temp = n;
    while (temp != 0) {
    digit = temp % 10;
    fact = 1;
    for (int i = 1; i <= digit; i++) {
    fact = fact * i;
    }
    sum = sum + fact;
    temp = temp / 10;
    }
    if (sum == n) {
        printf("%d is Strong.\n", n);
    }
    else {
        printf("%d is not Strong.\n", n);
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n, temp, sum = 0, digit, fact;
    cout << "Enter a number: ";
    cin >> n;
    temp = n;
    while (temp != 0) {
    digit = temp % 10;
    fact = 1;
    for (int i = 1; i <= digit; i++) {
    fact = fact * i;
    }
    sum = sum + fact;
    temp = temp / 10;
    }
    if (sum == n) {
        cout << n << " is a Strong number." << end
l;
    }
    else {
        cout << n << " is not a Strong number." <<
endl;
    }
    return 0;
}
```

২২. গ.সা.গু এবং ল.সা.গু 22. GCD AND LCM

```
গ.সা.গু (GCD) — Greatest Common Divisor:
  দুটি সংখ্যার সবচেয়ে বড় সাধারণ গুণনীয়ক।
  GCD(12, 8): 12 = 2²×3,  8 = 2³  →  GCD = 4

ল.সা.গু (LCM) — Least Common Multiple:
  দুটি সংখ্যার সবচেয়ে ছোট সাধারণ গুণিতক।
  LCM = (a × b) ÷ GCD(a, b)
  LCM(12, 8) = (12 × 8) ÷ 4 = 24

Euclidean Algorithm দিয়ে GCD:
  GCD(12, 8) → GCD(8, 4) → GCD(4, 0) → 4
```

C

```c
#include <stdio.h>
int main() {
    int a, b, x, y, gcd;
    int lcm;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);
    x = a;
    y = b;
    while (y != 0) {
    int t = y;
    y = x % y;
    x = t;
    }
    gcd = x;
    lcm = a / gcd * b;
    printf("GCD = %d\n", gcd);
    printf("LCM = %d\n", lcm);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int a, b, x, y, gcd;
    int lcm;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    x = a;
    y = b;
    while (y != 0) {
    int t = y;
    y = x % y;
    x = t;
    }
    gcd = x;
    lcm = a / gcd * b;
    cout << "GCD = " << gcd << endl;
    cout << "LCM = " << lcm << endl;
    return 0;
}
```

§ SERIES & SEQUENCES

২৩. ফ্যাক্টরিয়াল 23. FACTORIAL

```
ফ্যাক্টরিয়াল (!) কী?
  n! = n × (n-1) × (n-2) × ... × 2 × 1
  0! = 1  (সংজ্ঞা অনুযায়ী)
  1! = 1
  5! = 5 × 4 × 3 × 2 × 1 = 120
  10! = 3,628,800

সতর্কতা: সংখ্যা দ্রুত বড় হয়। 13! এর বেশি int-এ ধরে না, long long ব্যবহার করতে হয়।
```

C

```c
#include <stdio.h>
int main() {
    int n;
    long long fact = 1;
    printf("Enter a number: ");
    scanf("%d", &n);
    if (n < 0) {
        printf("Error: factorial of negative number is undefine
d.\n");
        return 1;
    }
    for (int i = 1; i <= n; i++) {
    fact = fact * i;
    }
    printf("%d! = %lld\n", n, fact);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    long long fact = 1;
    cout << "Enter a number: ";
    cin >> n;
    if (n < 0) {
        cout << "Error: factorial of negative number is undefine
d." << endl;
        return 1;
    }
    for (int i = 1; i <= n; i++) {
    fact = fact * i;
    }
    cout << n << "! = " << fact << endl;
    return 0;
}
```

২৪. সিরিজ: ১! + ২! + ... + n! 24. SERIES: 1! + 2! + ... + n!

```
সিরিজ: ১! + ২! + ... + n!

n=5 হলে:
1! + 2! + 3! + 4! + 5!
= 1 + 2 + 6 + 24 + 120
= 153
```

C

```c
#include <stdio.h>
int main() {
    int n;
    long long sum = 0, fact = 1;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
    fact = fact * i;
    sum = sum + fact;
    }
    printf("Sum = %lld\n", sum);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    long long sum = 0, fact = 1;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
    fact = fact * i;
    sum = sum + fact;
    }
    cout << "1! + 2! + ... + " << n << "! = " << sum << endl;
    return 0;
}
```

২৫. সিরিজ: ২! + ৪! + ... + n! 25. SERIES: 2! + 4! + ... + n!

```
সিরিজ: ২! + ৪! + ... + n!

n=6 হলে (জোড় পদ):
2! + 4! + 6!
= 2 + 24 + 720
= 746
```

C

```c
#include <stdio.h>
int main() {
    int n;
    long long sum = 0, fact;
    printf("Enter n (even): ");
    scanf("%d", &n);
    for (int i = 2; i <= n; i = i + 2) {
    fact = 1;
    for (int j = 1; j <= i; j++) {
    fact = fact * j;
    }
    sum = sum + fact;
    }
    printf("Sum = %lld\n", sum);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    long long sum = 0, fact;
    cout << "Enter n (even number): ";
    cin >> n;
    for (int i = 2; i <= n; i = i + 2) {
    fact = 1;
    for (int j = 1; j <= i; j++) {
    fact = fact * j;
    }
    sum = sum + fact;
    }
    cout << "2! + 4! + ... + " << n << "! = " << sum << endl;
    return 0;
}
```

২৬. সিরিজ: ১! + ৩! + ... + n! 26. SERIES: 1! + 3! + ... + n!

```
সিরিজ: ১! + ৩! + ... + n!

n=5 হলে (বিজোড় পদ):
1! + 3! + 5!
= 1 + 6 + 120
= 127
```

C

```c
#include <stdio.h>
int main() {
    int n;
    long long sum = 0, fact;
    printf("Enter n (odd): ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i = i + 2) {
    fact = 1;
    for (int j = 1; j <= i; j++) {
    fact = fact * j;
    }
    sum = sum + fact;
    }
    printf("Sum = %lld\n", sum);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    long long sum = 0, fact;
    cout << "Enter n (odd number): ";
    cin >> n;
    for (int i = 1; i <= n; i = i + 2) {
    fact = 1;
    for (int j = 1; j <= i; j++) {
    fact = fact * j;
    }
    sum = sum + fact;
    }
    cout << "1! + 3! + ... + " << n << "! = " << sum << endl;
    return 0;
}
```

২৭ক. ফিবোনাচ্চি - পদগুলো প্রিন্ট 27a. FIBONACCI - PRINT TERMS

```
ফিবোনাচ্চি সিরিজ

প্রতিটি পদ = পূর্ববর্তী দুটি পদের যোগফল

n=8 হলে আউটপুট:
0  1  1  2  3  5  8  13
↑  ↑  ↑
0+1=1, 1+1=2, 1+2=3, 2+3=5 ...

গঠন: F(0)=0, F(1)=1
F(n) = F(n-1) + F(n-2)
```

C

```c
#include <stdio.h>
int main() {
    int n, a = 0, b = 1, c;
    printf("Enter number of terms: ");
    scanf("%d", &n);
    printf("Series: %d %d", a, b);
    for (int i = 3; i <= n; i++) {
    c = a + b;
    printf(" %d", c); a = b; b = c;
    }
    printf("\n");
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n, a = 0, b = 1, c;
    cout << "Enter number of terms: ";
    cin >> n;
    cout << "Series: " << a << " " << b;
    for (int i = 3; i <= n; i++) {
    c = a + b;
    cout << " " << c; a = b; b = c;
    }
    cout << endl;
    return 0;
}
```

২৭খ. ফিবোনাচ্চি - প্রিন্ট ও যোগফল 27b. FIBONACCI - PRINT & SUM

```
ফিবোনাচ্চি সিরিজ ও যোগফল

n=6 হলে আউটপুট:
0  1  1  2  3  5
যোগফল = 0+1+1+2+3+5 = 12
```

C

```c
#include <stdio.h>
int main() {
    int n, a = 0, b = 1, c;
    int sum;
    printf("Enter number of terms: ");
    scanf("%d", &n);
    sum = a + b;
    printf("Series: %d %d", a, b);
    for (int i = 3; i <= n; i++) {
    c = a + b;
    printf(" %d", c);
    sum = sum + c;
    a = b;
    b = c;
    }
    printf("\nSum = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n, a = 0, b = 1, c;
    int sum;
    cout << "Enter number of terms: ";
    cin >> n;
    sum = a + b;
    cout << "Series: " << a << " " << b;
    for (int i = 3; i <= n; i++) {
    c = a + b;
    cout << " " << c;
    sum = sum + c;
    a = b;
    b = c;
    }
    cout << "\nSum = " << sum << endl;
    return 0;
}
```

২৮. সিরিজ: ১ + ২ + ... + n 28. SERIES: 1 + 2 + ... + n

```
সিরিজ: ১ + ২ + ... + n

n=5: 1 + 2 + 3 + 4 + 5 = 15
সূত্র: n(n+1)/2
```

C

```c
#include <stdio.h>
int main() {
    int n;
    int sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
    sum = sum + i;
    }
    printf("Sum = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    int sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
    sum = sum + i;
    }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

২৯. সিরিজ: ১ + ৩ + ৫ + ... + n 29. SERIES: 1 + 3 + 5 + ... + n

```
সিরিজ: ১ + ৩ + ৫ + ... + n

n=7 হলে (বিজোড় সংখ্যা):
1 + 3 + 5 + 7 = 16
```

C

```c
#include <stdio.h>
int main() {
    int n;
    int sum = 0;
    printf("Enter n (odd): ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i = i + 2) {
    sum = sum + i;
    }
    printf("Sum = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    int sum = 0;
    cout << "Enter n (odd): ";
    cin >> n;
    for (int i = 1; i <= n; i = i + 2) {
    sum = sum + i;
    }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩০. সিরিজ: ২ + ৪ + ৬ + ... + n 30. SERIES: 2 + 4 + 6 + ... + n

```
সিরিজ: ২ + ৪ + ৬ + ... + n

n=6 হলে (জোড় সংখ্যা):
2 + 4 + 6 = 12
```

C

```c
#include <stdio.h>
int main() {
    int n;
    int sum = 0;
    printf("Enter n (even): ");
    scanf("%d", &n);
    for (int i = 2; i <= n; i = i + 2) {
    sum = sum + i;
    }
    printf("Sum = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    int sum = 0;
    cout << "Enter n (even): ";
    cin >> n;
    for (int i = 2; i <= n; i = i + 2) {
    sum = sum + i;
    }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩১. সিরিজ: ১^১ + ২^২ + ... + n^n 31. SERIES: 1^1 + 2^2 + ... + n^n

```
সিরিজ: ১^১ + ২^২ + ... + n^n

n=4: 1¹ + 2² + 3³ + 4
= 1 + 4 + 27 + 256 = 288
```

C

```c
#include <stdio.h>
int main() {
    int n;
    int sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
    int p = 1;
    for (int j = 0; j < i; j++) {
    p = p * i;
    }
    sum = sum + p;
    }
    printf("Sum = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    int sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
    int p = 1;
    for (int j = 0; j < i; j++) {
    p = p * i;
    }
    sum = sum + p;
    }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩২. সিরিজ: ১^২ + ২^২ + ... + n^2 32. SERIES: 1^2 + 2^2 + ... + n^2

```
সিরিজ: ১² + ২² + ... + n²

n=4: 1² + 2² + 3² + 4²
= 1 + 4 + 9 + 16 = 30
```

C

```c
#include <stdio.h>
int main() {
    int n;
    int sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
    sum = sum + i * i;
    }
    printf("Sum = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    int sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
    sum = sum + i * i;
    }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩৩. সিরিজ: ১^২ + ৩^২ + ... + n^2 33. SERIES: 1^2 + 3^2 + ... + n^2

```
সিরিজ: ১² + ৩² + ... + n² (বিজোড়)

n=5: 1² + 3² + 5²
= 1 + 9 + 25 = 35
```

C

```c
#include <stdio.h>
int main() {
    int n;
    int sum = 0;
    printf("Enter n (odd): ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i = i + 2) {
    sum = sum + i * i;
    }
    printf("Sum = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    int sum = 0;
    cout << "Enter n (odd): ";
    cin >> n;
    for (int i = 1; i <= n; i = i + 2) {
    sum = sum + i * i;
    }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩৪. সিরিজ: ২^২ + ৪^২ + ... + n^2 34. SERIES: 2^2 + 4^2 + ... + n^2

```
সিরিজ: ২² + ৪² + ... + n² (জোড়)

n=6: 2² + 4² + 6²
= 4 + 16 + 36 = 56
```

C

```c
#include <stdio.h>
int main() {
    int n;
    int sum = 0;
    printf("Enter n (even): ");
    scanf("%d", &n);
    for (int i = 2; i <= n; i = i + 2) {
    sum = sum + i * i;
    }
    printf("Sum = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    int sum = 0;
    cout << "Enter n (even): ";
    cin >> n;
    for (int i = 2; i <= n; i = i + 2) {
    sum = sum + i * i;
    }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩৫. সিরিজ: ১ + ২ + ৪ + ৮ + ... 35. SERIES: 1 + 2 + 4 + 8 + ...

```
সিরিজ: ১ + ২ + ৪ + ৮ + ...

n=5 পদ: 1 + 2 + 4 + 8 + 16 = 31
(প্রতিটি পদ = আগেরটি × 2)
```

C

```c
#include <stdio.h>
int main() {
    int n;
    int sum = 0, term = 1;
    printf("Enter number of terms: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
    sum = sum + term;
    term = term * 2;
    }
    printf("Sum = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    int sum = 0, term = 1;
    cout << "Enter number of terms: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
    sum = sum + term;
    term = term * 2;
    }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩৬. সিরিজ: ১ + ৩ + ৯ + ২৭ + ... 36. SERIES: 1 + 3 + 9 + 27 + ...

```
সিরিজ: ১ + ৩ + ৯ + ২৭ + ...

n=4 পদ: 1 + 3 + 9 + 27 = 40
(প্রতিটি পদ = আগেরটি × 3)
```

C

```c
#include <stdio.h>
int main() {
    int n;
    int sum = 0, term = 1;
    printf("Enter number of terms: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
    sum = sum + term;
    term = term * 3;
    }
    printf("Sum = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    int sum = 0, term = 1;
    cout << "Enter number of terms: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
    sum = sum + term;
    term = term * 3;
    }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩৭. সিরিজ: ১/১ + ১/২ + ... + ১/n 37. SERIES: 1/1 + 1/2 + ... + 1/n

```
সিরিজ: ১/১ + ১/২ + ... + ১/n

n=4: 1/1 + 1/2 + 1/3 + 1/4
= 1 + 0.5 + 0.333 + 0.25
≈ 2.083
```

C

```c
#include <stdio.h>
int main() {
    int n;
    float sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
    sum = sum + 1.0 / i;
    }
    printf("Sum = %.4f\n", sum);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    float sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
    sum = sum + 1.0 / i;
    }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩৮. সিরিজ: ১/১^২ + ১/২^২ + ... 38. SERIES: 1/1^2 + 1/2^2 + ...

```
সিরিজ: ১/১² + ১/২² + ১/৩² + ...

n=4: 1/1 + 1/4 + 1/9 + 1/16
= 1 + 0.25 + 0.111 + 0.0625
≈ 1.424
```

C

```c
#include <stdio.h>
int main() {
    int n;
    float sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
    sum = sum + 1.0 / ((float)i * i);
    }
    printf("Sum = %.4f\n", sum);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    float sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++) {
    sum = sum + 1.0 / ((float)i * i);
    }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩৯. প্যাটার্ন - তারা বৃদ্ধি 39. PATTERN - STARS INCREASING

ভিজ্যুয়াল আউটপুট (n=5):

```
*
**
***
****
*****
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= r; c++) {
    printf("*");
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= row; col++) {
    cout << "*";
    }
    cout << endl;
    }
    return 0;
}
```

৪০. প্যাটার্ন - সংখ্যা বৃদ্ধি 40. PATTERN - NUMBERS INCREASING

ভিজ্যুয়াল আউটপুট (n=5):

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= r; c++) {
    printf("%d", c);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= row; col++) {
    cout << col;
    }
    cout << endl;
    }
    return 0;
}
```

৪১. প্যাটার্ন - সারি সংখ্যা পুনরাবৃত্তি 41. PATTERN - REPEATED ROW NUM

ভিজ্যুয়াল আউটপুট (n=5):

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= r; c++) {
    printf("%d", r);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= row; col++) {
    cout << row;
    }
    cout << endl;
    }
    return 0;
}
```

৪২. প্যাটার্ন - ১ ০ পর্যায়ক্রমে (১ দিয়ে শুরু) 42. PATTERN - 1 0 ALT START 1

ভিজ্যুয়াল আউটপুট (n=5):

```
1
1 0
1 0 1
1 0 1 0
1 0 1 0 1
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= r; c++) {
    printf("%d", (c % 2 == 1) ? 1 : 0);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= row; col++) {
    cout << (col % 2 == 1 ? 1 : 0);
    }
    cout << endl;
    }
    return 0;
}
```

৪৩. প্যাটার্ন - ১ ০ পর্যায়ক্রমে (বিজোড়=১ জোড়=০) 43. PATTERN - 1 0 ALT ODD=1 EVEN=0

ভিজ্যুয়াল আউটপুট (n=5):

```
1
0 0
1 1 1
0 0 0 0
1 1 1 1 1
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    int v = (r % 2 == 1) ? 1 : 0;
    for (int c = 1; c <= r; c++) {
    printf("%d", v);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    int val = (row % 2 == 1) ? 1 : 0;
    for (int col = 1; col <= row; col++) {
    cout << val;
    }
    cout << endl;
    }
    return 0;
}
```

৪৪. প্যাটার্ন - বর্ণমালা পুনরাবৃত্তি 44. PATTERN - REPEATED ALPHABET

ভিজ্যুয়াল আউটপুট (n=5):

```
A
B B
C C C
D D D D
E E E E E
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= r; c++) {
    printf("%c", 'A' + r - 1);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= row; col++) {
    cout << (char)('A' + row - 1);
    }
    cout << endl;
    }
    return 0;
}
```

৪৫. প্যাটার্ন - বর্ণমালা ক্রমবর্ধমান 45. PATTERN - ALPHABETS INC

ভিজ্যুয়াল আউটপুট (n=5):

```
A
A B
A B C
A B C D
A B C D E
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= r; c++) {
    printf("%c", 'A' + c - 1);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= row; col++) {
    cout << (char)('A' + col - 1);
    }
    cout << endl;
    }
    return 0;
}
```

§ PATTERNS — DECREASING

৪৬. প্যাটার্ন - তারা হ্রাস 46. PATTERN - STARS DECREASING

ভিজ্যুয়াল আউটপুট (n=5):

```
*****
****
***
**
*
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
    for (int c = 1; c <= r; c++) {
    printf("*");
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
    for (int col = 1; col <= row; col++) {
    cout << "*";
    }
    cout << endl;
    }
    return 0;
}
```

৪৭. প্যাটার্ন - সংখ্যা হ্রাস 47. PATTERN - NUMBERS DECREASING

ভিজ্যুয়াল আউটপুট (n=5):

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
    for (int c = 1; c <= r; c++) {
    printf("%d", c);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
    for (int col = 1; col <= row; col++) {
    cout << col;
    }
    cout << endl;
    }
    return 0;
}
```

৪৮. প্যাটার্ন - সারি সংখ্যা পুনরাবৃত্তি হ্রাস 48. PATTERN - REP ROW DECREASING

ভিজ্যুয়াল আউটপুট (n=5):

```
5 5 5 5 5
4 4 4 4
3 3 3
2 2
1
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
    for (int c = 1; c <= r; c++) {
    printf("%d", r);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
    for (int col = 1; col <= row; col++) {
    cout << row;
    }
    cout << endl;
    }
    return 0;
}
```

৪৯. প্যাটার্ন - ১ ০ পর্যায়ক্রমে হ্রাস 49. PATTERN - 1 0 ALT DECREASING

ভিজ্যুয়াল আউটপুট (n=5):

```
1 0 1 0 1
1 0 1 0
1 0 1
1 0
1
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
    for (int c = 1; c <= r; c++) {
    printf("%d", (c % 2 == 1) ? 1 : 0);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
    for (int col = 1; col <= row; col++) {
    cout << (col % 2 == 1 ? 1 : 0);
    }
    cout << endl;
    }
    return 0;
}
```

৫০. প্যাটার্ন - সারি ১/০ হ্রাস 50. PATTERN - ROW 1/0 DECREASING

ভিজ্যুয়াল আউটপুট (n=5):

```
1 1 1 1 1
0 0 0 0
1 1 1
0 0
1
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
    int v = (r % 2 == 1) ? 1 : 0;
    for (int c = 1; c <= r; c++) {
    printf("%d", v);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
    int val = (row % 2 == 1) ? 1 : 0;
    for (int col = 1; col <= row; col++) {
    cout << val;
    }
    cout << endl;
    }
    return 0;
}
```

৫১. প্যাটার্ন - বর্ণমালা হ্রাসমান 51. PATTERN - ALPHABETS DEC

ভিজ্যুয়াল আউটপুট (n=5):

```
A B C D E
A B C D
A B C
A B
A
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
    for (int c = 1; c <= r; c++) {
    printf("%c", 'A' + c - 1);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
    for (int col = 1; col <= row; col++) {
    cout << (char)('A' + col - 1);
    }
    cout << endl;
    }
    return 0;
}
```

৫২. প্যাটার্ন - বর্ণমালা পুনরাবৃত্তি হ্রাস 52. PATTERN - REP ALPHABET DEC

ভিজ্যুয়াল আউটপুট (n=5):

```
E E E E E
D D D D
C C C
B B
A
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
    for (int c = 1; c <= r; c++) {
    printf("%c", 'A' + r - 1);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
    for (int col = 1; col <= row; col++) {
    cout << (char)('A' + row - 1);
    }
    cout << endl;
    }
    return 0;
}
```

৫৩. প্যাটার্ন - তারা ডায়মন্ড 53. PATTERN - STARS DIAMOND

ভিজ্যুয়াল আউটপুট (n=5): ◄── middle row

```
*
**
*****
*******
*********
*******
*****
**
*
```

▲ INCREASING (TOP HALF) · ★ MIDDLE ROW · DECREASING (BOTTOM HALF) ▼

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= r; c++) {
    printf("*");
    }
    printf("\n");
    }
    for (int r = n - 1; r >= 1; r--) {
    for (int c = 1; c <= r; c++) {
    printf("*");
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= row; col++) {
    cout << "*";
    }
    cout << endl;
    }
    for (int row = n - 1; row >= 1; row--) {
    for (int col = 1; col <= row; col++) {
    cout << "*";
    }
    cout << endl;
    }
    return 0;
}
```

৫৪. প্যাটার্ন - সংখ্যা ডায়মন্ড 54. PATTERN - NUMBERS DIAMOND

ভিজ্যুয়াল আউটপুট (n=4): ◄── middle row

```
1
1 2
1 2 3
1 2 3 4
1 2 3
1 2
1
```

▲ INCREASING (TOP HALF) · ★ MIDDLE ROW · DECREASING (BOTTOM HALF) ▼

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= r; c++) {
    printf("%d", c);
    }
    printf("\n");
    }
    for (int r = n - 1; r >= 1; r--) {
    for (int c = 1; c <= r; c++) {
    printf("%d", c);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= row; col++) {
    cout << col;
    }
    cout << endl;
    }
    for (int row = n - 1; row >= 1; row--) {
    for (int col = 1; col <= row; col++) {
    cout << col;
    }
    cout << endl;
    }
    return 0;
}
```

৫৫. প্যাটার্ন - সারি সংখ্যা পুনরাবৃত্তি ডায়মন্ড 55. PATTERN - REP ROW NUM DIAMOND

ভিজ্যুয়াল আউটপুট (n=4): ◄── middle row

```
1
2 2
3 3 3
4 4 4 4
3 3 3
2 2
1
```

▲ INCREASING (TOP HALF) · ★ MIDDLE ROW · DECREASING (BOTTOM HALF) ▼

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= r; c++) {
    printf("%d", r);
    }
    printf("\n");
    }
    for (int r = n - 1; r >= 1; r--) {
    for (int c = 1; c <= r; c++) {
    printf("%d", r);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= row; col++) {
    cout << row;
    }
    cout << endl;
    }
    for (int row = n - 1; row >= 1; row--) {
    for (int col = 1; col <= row; col++) {
    cout << row;
    }
    cout << endl;
    }
    return 0;
}
```

৫৬. প্যাটার্ন - ১/০ (১ দিয়ে শুরু) ডায়মন্ড 56. PATTERN - 1/0 START 1 DIAMOND

ভিজ্যুয়াল আউটপুট (n=4): ◄── middle row

```
1
1 0
1 0 1
1 0 1 0
1 0 1
1 0
1
```

▲ INCREASING (TOP HALF) · ★ MIDDLE ROW · DECREASING (BOTTOM HALF) ▼

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= r; c++) {
    printf("%d", (c % 2 == 1) ? 1 : 0);
    }
    printf("\n");
    }
    for (int r = n - 1; r >= 1; r--) {
    for (int c = 1; c <= r; c++) {
    printf("%d", (c % 2 == 1) ? 1 : 0);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= row; col++) {
    cout << (col % 2 == 1 ? 1 : 0);
    }
    cout << endl;
    }
    for (int row = n - 1; row >= 1; row--) {
    for (int col = 1; col <= row; col++) {
    cout << (col % 2 == 1 ? 1 : 0);
    }
    cout << endl;
    }
    return 0;
}
```

৫৭. প্যাটার্ন - সারি ১/০ ডায়মন্ড 57. PATTERN - ROW 1/0 DIAMOND

ভিজ্যুয়াল আউটপুট (n=4): ◄── middle row

```
1
0 0
1 1 1
0 0 0 0
1 1 1
0 0
1
```

▲ INCREASING (TOP HALF) · ★ MIDDLE ROW · DECREASING (BOTTOM HALF) ▼

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    int v = (r % 2 == 1) ? 1 : 0;
    for (int c = 1; c <= r; c++) {
    printf("%d", v);
    }
    printf("\n");
    }
    for (int r = n - 1; r >= 1; r--) {
    int v = (r % 2 == 1) ? 1 : 0;
    for (int c = 1; c <= r; c++) {
    printf("%d", v);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    int val = (row % 2 == 1) ? 1 : 0;
    for (int col = 1; col <= row; col++) {
    cout << val;
    }
    cout << endl;
    }
    for (int row = n - 1; row >= 1; row--) {
    int val = (row % 2 == 1) ? 1 : 0;
    for (int col = 1; col <= row; col++) {
    cout << val;
    }
    cout << endl;
    }
    return 0;
}
```

৫৮. প্যাটার্ন - বর্ণমালা পুনরাবৃত্তি ডায়মন্ড 58. PATTERN - REP ALPHABET DIAMOND

ভিজ্যুয়াল আউটপুট (n=4): ◄── middle row

```
A
B B
C C C
D D D D
C C C
B B
A
```

▲ INCREASING (TOP HALF) · ★ MIDDLE ROW · DECREASING (BOTTOM HALF) ▼

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= r; c++) {
    printf("%c", 'A' + r - 1);
    }
    printf("\n");
    }
    for (int r = n - 1; r >= 1; r--) {
    for (int c = 1; c <= r; c++) {
    printf("%c", 'A' + r - 1);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= row; col++) {
    cout << (char)('A' + row - 1);
    }
    cout << endl;
    }
    for (int row = n - 1; row >= 1; row--) {
    for (int col = 1; col <= row; col++) {
    cout << (char)('A' + row - 1);
    }
    cout << endl;
    }
    return 0;
}
```

৫৯. প্যাটার্ন - বর্ণমালা ক্রম ডায়মন্ড 59. PATTERN - ALPHABET SEQ DIAMOND

ভিজ্যুয়াল আউটপুট (n=4): ◄── middle row

```
A
A B
A B C
A B C D
A B C
A B
A
```

▲ INCREASING (TOP HALF) · ★ MIDDLE ROW · DECREASING (BOTTOM HALF) ▼

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= r; c++) {
    printf("%c", 'A' + c - 1);
    }
    printf("\n");
    }
    for (int r = n - 1; r >= 1; r--) {
    for (int c = 1; c <= r; c++) {
    printf("%c", 'A' + c - 1);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= row; col++) {
    cout << (char)('A' + col - 1);
    }
    cout << endl;
    }
    for (int row = n - 1; row >= 1; row--) {
    for (int col = 1; col <= row; col++) {
    cout << (char)('A' + col - 1);
    }
    cout << endl;
    }
    return 0;
}
```

§ PATTERNS — RIGHT-ALIGNED

৬০. প্যাটার্ন - ফাঁকা বর্গ 60. PATTERN - HOLLOW SQUARE

ভিজ্যুয়াল আউটপুট (n=5):

```
* * * * *
*       *
*       *
*       *
* * * * *
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= n; c++) {
    if (r == 1 || r == n || c == 1 || c == n)
    printf("*");
    else
    printf(" ");
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= n; col++) {
    if (row == 1 || row == n || col == 1 || col == n)
    cout << "*";
    else
    cout << " ";
    }
    cout << endl;
    }
    return 0;
}
```

৬১. প্যাটার্ন - ডানদিকে সাজানো তারা 61. PATTERN - RIGHT-ALIGNED STARS

ভিজ্যুয়াল আউটপুট (n=5):

```
*
* *
* * *
* * * *
* * * * *
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int s = 1; s <= n - r; s++) {
    printf(" ");
    }
    for (int c = 1; c <= r; c++) {
    printf("*");
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int sp = 1; sp <= n - row; sp++) {
    cout << " ";
    }
    for (int col = 1; col <= row; col++) {
    cout << "*";
    }
    cout << endl;
    }
    return 0;
}
```

৬২. প্যাটার্ন - ডানদিকে সাজানো সংখ্যা 62. PATTERN - RIGHT-ALIGNED NUMBERS

ভিজ্যুয়াল আউটপুট (n=5):

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int s = 1; s <= n - r; s++) {
    printf(" ");
    }
    for (int c = 1; c <= r; c++) {
    printf("%d", c);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int sp = 1; sp <= n - row; sp++) {
    cout << " ";
    }
    for (int col = 1; col <= row; col++) {
    cout << col;
    }
    cout << endl;
    }
    return 0;
}
```

৬৩. প্যাটার্ন - ডানদিকে সাজানো সারি পুনরাবৃত্তি 63. PATTERN - RIGHT-ALIGNED REP ROW

ভিজ্যুয়াল আউটপুট (n=5):

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int s = 1; s <= n - r; s++) {
    printf(" ");
    }
    for (int c = 1; c <= r; c++) {
    printf("%d", r);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int sp = 1; sp <= n - row; sp++) {
    cout << " ";
    }
    for (int col = 1; col <= row; col++) {
    cout << row;
    }
    cout << endl;
    }
    return 0;
}
```

৬৪. প্যাটার্ন - ডানদিকে সাজানো ১/০ (১ দিয়ে শুরু) 64. PATTERN - RIGHT-ALIGNED 1/0 START 1

ভিজ্যুয়াল আউটপুট (n=5):

```
1
1 0
1 0 1
1 0 1 0
1 0 1 0 1
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int s = 1; s <= n - r; s++) {
    printf(" ");
    }
    for (int c = 1; c <= r; c++) {
    printf("%d", (c % 2 == 1) ? 1 : 0);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int sp = 1; sp <= n - row; sp++) {
    cout << " ";
    }
    for (int col = 1; col <= row; col++) {
    cout << (col % 2 == 1 ? 1 : 0);
    }
    cout << endl;
    }
    return 0;
}
```

৬৫. প্যাটার্ন - ডানদিকে সাজানো সারি ১/০ 65. PATTERN - RIGHT-ALIGNED ROW 1/0

ভিজ্যুয়াল আউটপুট (n=5):

```
1
0 0
1 1 1
0 0 0 0
1 1 1 1 1
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    int v = (r % 2 == 1) ? 1 : 0;
    for (int s = 1; s <= n - r; s++) {
    printf(" ");
    }
    for (int c = 1; c <= r; c++) {
    printf("%d", v);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    int val = (row % 2 == 1) ? 1 : 0;
    for (int sp = 1; sp <= n - row; sp++) {
    cout << " ";
    }
    for (int col = 1; col <= row; col++) {
    cout << val;
    }
    cout << endl;
    }
    return 0;
}
```

৬৬. প্যাটার্ন - ডানদিকে সাজানো বর্ণমালা পুনরাবৃত্তি 66. PATTERN - RIGHT-ALIGNED REP ALPHA

ভিজ্যুয়াল আউটপুট (n=5):

```
A
B B
C C C
D D D D
E E E E E
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int s = 1; s <= n - r; s++) {
    printf(" ");
    }
    for (int c = 1; c <= r; c++) {
    printf("%c", 'A' + r - 1);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int sp = 1; sp <= n - row; sp++) {
    cout << " ";
    }
    for (int col = 1; col <= row; col++) {
    cout << (char)('A' + row - 1);
    }
    cout << endl;
    }
    return 0;
}
```

৬৭. প্যাটার্ন - ডানদিকে সাজানো বর্ণমালা ক্রম 67. PATTERN - RIGHT-ALIGNED ALPHA SEQ

ভিজ্যুয়াল আউটপুট (n=5):

```
A
A B
A B C
A B C D
A B C D E
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int s = 1; s <= n - r; s++) {
    printf(" ");
    }
    for (int c = 1; c <= r; c++) {
    printf("%c", 'A' + c - 1);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int sp = 1; sp <= n - row; sp++) {
    cout << " ";
    }
    for (int col = 1; col <= row; col++) {
    cout << (char)('A' + col - 1);
    }
    cout << endl;
    }
    return 0;
}
```

৬৮. প্যাটার্ন - বামদিকে ফাঁকা তারা হ্রাস 68. PATTERN - LEFT-INDENT STARS DEC

ভিজ্যুয়াল আউটপুট (n=5):

```
* * * * *
* * * *
* * *
* *
*
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
    for (int s = 1; s <= n - r; s++) {
    printf(" ");
    }
    for (int c = 1; c <= r; c++) {
    printf("*");
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
    for (int sp = 1; sp <= n - row; sp++) {
    cout << " ";
    }
    for (int col = 1; col <= row; col++) {
    cout << "*";
    }
    cout << endl;
    }
    return 0;
}
```

৬৯. প্যাটার্ন - বামদিকে ফাঁকা সংখ্যা হ্রাস 69. PATTERN - LEFT-INDENT NUMBERS DEC

ভিজ্যুয়াল আউটপুট (n=5):

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
    for (int s = 1; s <= n - r; s++) {
    printf(" ");
    }
    for (int c = 1; c <= r; c++) {
    printf("%d", c);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
    for (int sp = 1; sp <= n - row; sp++) {
    cout << " ";
    }
    for (int col = 1; col <= row; col++) {
    cout << col;
    }
    cout << endl;
    }
    return 0;
}
```

৭০. প্যাটার্ন - বামদিকে ফাঁকা সারি পুনরাবৃত্তি হ্রাস 70. PATTERN - LEFT-INDENT REP ROW DEC

ভিজ্যুয়াল আউটপুট (n=5):

```
5 5 5 5 5
4 4 4 4
3 3 3
2 2
1
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
    for (int s = 1; s <= n - r; s++) {
    printf(" ");
    }
    for (int c = 1; c <= r; c++) {
    printf("%d", r);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
    for (int sp = 1; sp <= n - row; sp++) {
    cout << " ";
    }
    for (int col = 1; col <= row; col++) {
    cout << row;
    }
    cout << endl;
    }
    return 0;
}
```

৭১. প্যাটার্ন - বামদিকে ফাঁকা ১/০ হ্রাস 71. PATTERN - LEFT-INDENT 1/0 DEC

ভিজ্যুয়াল আউটপুট (n=5):

```
1 0 1 0 1
1 0 1 0
1 0 1
1 0
1
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
    for (int s = 1; s <= n - r; s++) {
    printf(" ");
    }
    for (int c = 1; c <= r; c++) {
    printf("%d", (c % 2 == 1) ? 1 : 0);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
    for (int sp = 1; sp <= n - row; sp++) {
    cout << " ";
    }
    for (int col = 1; col <= row; col++) {
    cout << (col % 2 == 1 ? 1 : 0);
    }
    cout << endl;
    }
    return 0;
}
```

৭২. প্যাটার্ন - বামদিকে ফাঁকা সারি ১/০ হ্রাস 72. PATTERN - LEFT-INDENT ROW 1/0 DEC

ভিজ্যুয়াল আউটপুট (n=5):

```
1 1 1 1 1
0 0 0 0
1 1 1
0 0
1
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
    int v = (r % 2 == 1) ? 1 : 0;
    for (int s = 1; s <= n - r; s++) {
    printf(" ");
    }
    for (int c = 1; c <= r; c++) {
    printf("%d", v);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
    int val = (row % 2 == 1) ? 1 : 0;
    for (int sp = 1; sp <= n - row; sp++) {
    cout << " ";
    }
    for (int col = 1; col <= row; col++) {
    cout << val;
    }
    cout << endl;
    }
    return 0;
}
```

৭৩. প্যাটার্ন - বামদিকে ফাঁকা বর্ণমালা পুনরাবৃত্তি হ্রাস 73. PATTERN - LEFT-INDENT REP ALPHA DEC

ভিজ্যুয়াল আউটপুট (n=5):

```
E E E E E
D D D D
C C C
B B
A
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
    for (int s = 1; s <= n - r; s++) {
    printf(" ");
    }
    for (int c = 1; c <= r; c++) {
    printf("%c", 'A' + r - 1);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
    for (int sp = 1; sp <= n - row; sp++) {
    cout << " ";
    }
    for (int col = 1; col <= row; col++) {
    cout << (char)('A' + row - 1);
    }
    cout << endl;
    }
    return 0;
}
```

৭৪. প্যাটার্ন - বামদিকে ফাঁকা বর্ণমালা ক্রম হ্রাস 74. PATTERN - LEFT-INDENT ALPHA SEQ DEC

ভিজ্যুয়াল আউটপুট (n=5):

```
A B C D E
A B C D
A B C
A B
A
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = n; r >= 1; r--) {
    for (int s = 1; s <= n - r; s++) {
    printf(" ");
    }
    for (int c = 1; c <= r; c++) {
    printf("%c", 'A' + c - 1);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = n; row >= 1; row--) {
    for (int sp = 1; sp <= n - row; sp++) {
    cout << " ";
    }
    for (int col = 1; col <= row; col++) {
    cout << (char)('A' + col - 1);
    }
    cout << endl;
    }
    return 0;
}
```

৭৫. প্যাটার্ন - পূর্ণ তারা বর্গ 75. PATTERN - SOLID STAR SQUARE

ভিজ্যুয়াল আউটপুট (n=5):

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= n; c++) {
    printf("*");
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= n; col++) {
    cout << "*";
    }
    cout << endl;
    }
    return 0;
}
```

৭৬. প্যাটার্ন - সংখ্যা ক্রম বর্গ 76. PATTERN - NUMBER SEQUENCE SQUARE

ভিজ্যুয়াল আউটপুট (n=4):

```
1  2  3  4
5  6  7  8
9 10 11 12
13 14 15 16
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= n; c++) {
    printf("%d", c);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= n; col++) {
    cout << col;
    }
    cout << endl;
    }
    return 0;
}
```

৭৭. প্যাটার্ন - সারি সংখ্যা বর্গ 77. PATTERN - ROW NUMBER SQUARE

ভিজ্যুয়াল আউটপুট (n=4):

```
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= n; c++) {
    printf("%d", r);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= n; col++) {
    cout << row;
    }
    cout << endl;
    }
    return 0;
}
```

৭৮. প্যাটার্ন - ১০১০ কলাম পর্যায়ক্রমে বর্গ 78. PATTERN - 1010 COL ALT SQUARE

ভিজ্যুয়াল আউটপুট (n=4):

```
1 0 1 0
1 0 1 0
1 0 1 0
1 0 1 0
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= n; c++) {
    printf("%d", (c % 2 == 1) ? 1 : 0);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= n; col++) {
    cout << (col % 2 == 1 ? 1 : 0);
    }
    cout << endl;
    }
    return 0;
}
```

৭৯. প্যাটার্ন - ১১১১/০০০০ সারি পর্যায়ক্রমে বর্গ 79. PATTERN - 1111/0000 ROW ALT SQUARE

ভিজ্যুয়াল আউটপুট (n=4):

```
1 1 1 1
0 0 0 0
1 1 1 1
0 0 0 0
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    int v = (r % 2 == 1) ? 1 : 0;
    for (int c = 1; c <= n; c++) {
    printf("%d", v);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    int val = (row % 2 == 1) ? 1 : 0;
    for (int col = 1; col <= n; col++) {
    cout << val;
    }
    cout << endl;
    }
    return 0;
}
```

৮০. প্যাটার্ন - বর্ণমালা পুনরাবৃত্তি বর্গ 80. PATTERN - REPEATED ALPHABET SQUARE

ভিজ্যুয়াল আউটপুট (n=4):

```
A A A A
B B B B
C C C C
D D D D
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= n; c++) {
    printf("%c", 'A' + r - 1);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= n; col++) {
    cout << (char)('A' + row - 1);
    }
    cout << endl;
    }
    return 0;
}
```

৮১. প্যাটার্ন - বর্ণমালা ক্রম বর্গ 81. PATTERN - ALPHABET SEQ SQUARE

ভিজ্যুয়াল আউটপুট (n=4):

```
A B C D
A B C D
A B C D
A B C D
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= n; c++) {
    printf("%c", 'A' + c - 1);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= n; col++) {
    cout << (char)('A' + col - 1);
    }
    cout << endl;
    }
    return 0;
}
```

৮২. প্যাটার্ন - ক্রমিক সংখ্যা ত্রিভুজ 82. PATTERN - SEQUENTIAL NUMS TRIANGLE

ভিজ্যুয়াল আউটপুট (n=4):

```
1
2  3
4  5  6
7  8  9 10
```

C

```c
#include <stdio.h>
int main() {
    int n, count = 1;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= r; c++) {
    printf("%d ", count++);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n, count = 1;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= row; col++) {
    cout << count++ << " ";
    }
    cout << endl;
    }
    return 0;
}
```

৮৩. প্যাটার্ন - ০ থেকে ক্রমিক 83. PATTERN - SEQUENTIAL FROM 0

ভিজ্যুয়াল আউটপুট (n=4):

```
0
1  2
3  4  5
6  7  8  9
```

C

```c
#include <stdio.h>
int main() {
    int n, count = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= r; c++) {
    printf("%d ", count++);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n, count = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= row; col++) {
    cout << count++ << " ";
    }
    cout << endl;
    }
    return 0;
}
```

৮৪. প্যাটার্ন - গুণ ত্রিভুজ 84. PATTERN - MULTIPLICATION TRIANGLE

ভিজ্যুয়াল আউটপুট (n=4):

```
1
2  4
3  6  9
4  8 12 16
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int c = 1; c <= r; c++) {
    printf("%d ", r * c);
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int col = 1; col <= row; col++) {
    cout << row * col << " ";
    }
    cout << endl;
    }
    return 0;
}
```

৮৫. প্যাটার্ন - তারার পিরামিড 85. PATTERN - PYRAMID OF STARS

ভিজ্যুয়াল আউটপুট (n=5):

```
*
***
*****
*******
*********
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int r = 1; r <= n; r++) {
    for (int s = 1; s <= n - r; s++) {
    printf(" ");
    }
    for (int c = 1; c <= (2 * r - 1); c++) {
    printf("*");
    }
    printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    for (int row = 1; row <= n; row++) {
    for (int sp = 1; sp <= n - row; sp++) {
    cout << " ";
    }
    for (int col = 1; col <= (2 * row - 1); col++) {
    cout << "*";
    }
    cout << endl;
    }
    return 0;
}
```

৮৬. রিকার্শন দিয়ে ফ্যাক্টরিয়াল 86. FACTORIAL USING RECURSION

রিকার্শন দিয়ে ফ্যাক্টরিয়াল

```
fact(5) কীভাবে কাজ করে:
fact(5) = 5 × fact(4)
= 5 × 4 × fact(3)
= 5 × 4 × 3 × fact(2)
= 5 × 4 × 3 × 2 × fact(1)
= 5 × 4 × 3 × 2 × 1 = 120
```

C

```c
#include <stdio.h>
long long fact(int n) {
    if (n <= 1) {
        return 1;
    }
    return n * fact(n - 1);
}
int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);
    printf("%d! = %lld\n", n, fact(n));
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
long long fact(int n) {
    if (n <= 1) {
        return 1;
    }
    return n * fact(n - 1);
}
int main() {
    int n;
    cout << "Enter a number: ";
    cin >> n;
    cout << n << "! = " << fact(n) << endl;
    return 0;
}
```

৮৭. রিকার্শন দিয়ে ফিবোনাচ্চি 87. FIBONACCI USING RECURSION

রিকার্শন দিয়ে ফিবোনাচ্চি

```
fib(5) কীভাবে কাজ করে:

fib(5)
/      \
fib(4)  fib(3)
/    \   /  \
fib(3) fib(2) fib(2) fib(1)
...

আউটপুট (n=7): 0 1 1 2 3 5 8
```

C

```c
#include <stdio.h>
int fib(int n) {
    if (n <= 1) {
        return n;
    }
    return fib(n - 1) + fib(n - 2);
}
int main() {
    int n;
    printf("Enter number of terms: ");
    scanf("%d", &n);
    printf("Series: ");
    for (int i = 0; i < n; i++) {
    printf("%d ", fib(i));
    }
    printf("\n");
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int fib(int n) {
    if (n <= 1) {
        return n;
    }
    return fib(n - 1) + fib(n - 2);
}
int main() {
    int n;
    cout << "Enter number of terms: ";
    cin >> n;
    cout << "Series: ";
    for (int i = 0; i < n; i++) {
    cout << fib(i) << " ";
    }
    cout << endl;
    return 0;
}
```

৮৮. রিকার্শন দিয়ে ঘাত নির্ণয় 88. CALCULATE POWER USING RECURSION

```
রিকার্শন দিয়ে ঘাত (Power) নির্ণয়:
  power(2, 5) কীভাবে কাজ করে:
  power(2,5) = 2 × power(2,4)
             = 2 × 2 × power(2,3)
             = 2 × 2 × 2 × power(2,2)
             = 2 × 2 × 2 × 2 × power(2,1)
             = 2 × 2 × 2 × 2 × 2 × power(2,0)
             = 2 × 2 × 2 × 2 × 2 × 1 = 32
  Base case: power(x, 0) = 1 — এখানে রিকার্শন থামে।
```

C

```c
#include <stdio.h>
long long power(int b, int e) {
    if (e == 0) {
        return 1;
    }
    return b * power(b, e - 1);
}
int main() {
    int base, exponent;
    printf("Enter base and exponent: ");
    scanf("%d %d", &base, &exponent);
    if (exponent < 0) {
        printf("Error: negative exponent not supported.\n");
        return 1;
    }
    printf("%d^%d = %lld\n", base, exponent, power(base, exponen
t));
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
long long power(int b, int e) {
    if (e == 0) {
        return 1;
    }
    return b * power(b, e - 1);
}
int main() {
    int base, exponent;
    cout << "Enter base and exponent: ";
    cin >> base >> exponent;
    if (exponent < 0) {
        cout << "Error: negative exponent not supported." << end
l;
        return 1;
    }
    cout << base << "^" << exponent << " = " << power(base, expo
nent) << endl;
    return 0;
}
```

৮৯ক. অ্যারে - সর্বোচ্চ মান খোঁজা 89a. ARRAY - FIND MAXIMUM

```
অ্যারে থেকে সর্বোচ্চ মান খোঁজা:
  অ্যারে: [5, 12, 3, 8, 20, 1]
  শুরু: maxVal = 5  (প্রথম উপাদান)
  12 > 5  → maxVal = 12
   3 < 12 → কোনো পরিবর্তন নেই
   8 < 12 → কোনো পরিবর্তন নেই
  20 > 12 → maxVal = 20
   1 < 20 → কোনো পরিবর্তন নেই
  ফলাফল: Maximum = 20
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("How many numbers? ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter %d numbers: ", n);
    for (int i = 0; i < n; i++) {
    scanf("%d", &arr[i]);
    }
    int maxVal = arr[0];
    for (int i = 1; i < n; i++)
    if (arr[i] > maxVal) {
    maxVal = arr[i];
    }
    printf("Maximum = %d\n", maxVal);
    return 0;
}
```

C++

```cpp
#include<iostream>
#include<vector>
using namespace std;
int main() {
    int n;
    cout << "How many numbers? ";
    cin >> n;
    vector<int> arr(n);
    cout << "Enter " << n << " numbers: ";
    for (int i = 0; i < n; i++) {
    cin >> arr[i];
    }
    int maxVal = arr[0];
    for (int i = 1; i < n; i++)
    if (arr[i] > maxVal) {
    maxVal = arr[i];
    }
    cout << "Maximum = " << maxVal << endl;
    return 0;
}
```

৮৯খ. অ্যারে - সর্বনিম্ন মান খোঁজা 89b. ARRAY - FIND MINIMUM

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("How many numbers? ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter %d numbers: ", n);
    for (int i = 0; i < n; i++) {
    scanf("%d", &arr[i]);
    }
    int minVal = arr[0];
    for (int i = 1; i < n; i++)
    if (arr[i] < minVal) {
    minVal = arr[i];
    }
    printf("Minimum = %d\n", minVal);
    return 0;
}
```

C++

```cpp
#include<iostream>
#include<vector>
using namespace std;
int main() {
    int n;
    cout << "How many numbers? ";
    cin >> n;
    vector<int> arr(n);
    cout << "Enter " << n << " numbers: ";
    for (int i = 0; i < n; i++) {
    cin >> arr[i];
    }
    int minVal = arr[0];
    for (int i = 1; i < n; i++)
    if (arr[i] < minVal) {
    minVal = arr[i];
    }
    cout << "Minimum = " << minVal << endl;
    return 0;
}
```

৯০. অ্যারে - সর্বোচ্চ ও সর্বনিম্ন উভয় 90. ARRAY - FIND BOTH MAX AND MIN

```
একই লুপে সর্বোচ্চ ও সর্বনিম্ন উভয় খোঁজা:
  অ্যারে: [5, 12, 3, 8, 20, 1]
  maxVal = 5, minVal = 5  (দুটোই প্রথম উপাদান দিয়ে শুরু)
  প্রতিটি উপাদানের সাথে দুটো তুলনা:
    উপাদান > maxVal → maxVal আপডেট
    উপাদান < minVal → minVal আপডেট
  ফলাফল: Maximum = 20,  Minimum = 1
```

C

```c
#include <stdio.h>
int main() {
    int n;
    printf("How many numbers? ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter %d numbers: ", n);
    for (int i = 0; i < n; i++) {
    scanf("%d", &arr[i]);
    }
    int maxVal = arr[0], minVal = arr[0];
    for (int i = 1; i < n; i++) {
    if (arr[i] > maxVal) {
    maxVal = arr[i];
    }
    if (arr[i] < minVal) {
    minVal = arr[i];
    }
    }
    printf("Maximum = %d\nMinimum = %d\n", maxVal, minVal);
    return 0;
}
```

C++

```cpp
#include<iostream>
#include<vector>
using namespace std;
int main() {
    int n;
    cout << "How many numbers? ";
    cin >> n;
    vector<int> arr(n);
    cout << "Enter " << n << " numbers: ";
    for (int i = 0; i < n; i++) {
    cin >> arr[i];
    }
    int maxVal = arr[0], minVal = arr[0];
    for (int i = 1; i < n; i++) {
    if (arr[i] > maxVal) {
    maxVal = arr[i];
    }
    if (arr[i] < minVal) {
    minVal = arr[i];
    }
    }
    cout << "Maximum = " << maxVal << endl;
    cout << "Minimum = " << minVal << endl;
    return 0;
}
```

৯১. টেম্প ভেরিয়েবল ছাড়া দুটি সংখ্যা অদলবদল 91. SWAP TWO NUMBERS WITHOUT TEMP

```
তৃতীয় চলক (temp) ছাড়া অদলবদল — গাণিতিক কৌশল:
  a = 5,  b = 3
  a = a + b = 8      (a-তে যোগফল রাখা হলো)
  b = a - b = 8-3=5  (পুরনো a পাওয়া গেল)
  a = a - b = 8-5=3  (পুরনো b পাওয়া গেল)
  ফলাফল: a = 3,  b = 5  ✓
```

C

```c
#include <stdio.h>
int main() {
    int a, b;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);
    printf("Before: a=%d, b=%d\n", a, b);
    a = a + b;
    b = a - b;
    a = a - b;
    printf("After:  a=%d, b=%d\n", a, b);
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    int a, b;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    cout << "Before: a = " << a << ", b = " << b << endl;
    a = a + b;
    b = a - b;
    a = a - b;
    cout << "After : a = " << a << ", b = " << b << endl;
    return 0;
}
```

§ LOOP CONTROL

৯২ক. continue-এর উদাহরণ 92a. CONTINUE EXAMPLE

```
continue কী করে?
লুপের ভেতরে continue পেলে সেই পুনরাবৃত্তির বাকি অংশ বাদ দিয়ে
সরাসরি পরের পুনরাবৃত্তিতে চলে যায়।
উদাহরণ (i=3 হলে skip): আউটপুট → 1 2 4 5 7 8 10  (3,6,9 বাদ)
```

C

```c
#include <stdio.h>
int main() {
    for (int i = 1; i <= 10; i++) {
    if (i % 3 == 0) {
    continue;
    }
    printf("%d ", i);
    }
    printf("\n");
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 10; i++) {
    if (i % 3 == 0) {
    continue;
    }
    cout << i << " ";
    }
    cout << endl;
    return 0;
}
```

৯২খ. break-এর উদাহরণ 92b. BREAK EXAMPLE

```
break কী করে?
লুপের ভেতরে break পেলে সম্পূর্ণ লুপ বন্ধ হয়ে বাইরে চলে আসে।
উদাহরণ (i==6 হলে stop): আউটপুট → 1 2 3 4 5  (6 আসতেই থামে)

continue vs break:
  continue → শুধু এই পুনরাবৃত্তি বাদ, লুপ চলতে থাকে।
  break    → পুরো লুপই বন্ধ।
```

C

```c
#include <stdio.h>
int main() {
    for (int i = 1; i <= 10; i++) {
    if (i == 6) {
    break;
    }
    printf("%d ", i);
    }
    printf("\n");
    return 0;
}
```

C++

```cpp
#include<iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 10; i++) {
    if (i == 6) {
    break;
    }
    cout << i << " ";
    }
    cout << endl;
    return 0;
}
```